

SMART IRRIGATION AND WATER MANAGEMENT SYSTEM FOR EFFICIENT AGRICULTURAL WATER USAGE

A PROJECT REPORT

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1. Project Overview

The Smart Irrigation and Water Management System is an automated agricultural system developed to provide water to crops based on their actual water requirements. Traditional irrigation methods often depend on manual operation or fixed watering schedules, which can result in water wastage, over-irrigation, under-irrigation, and increased labor requirements. This project aims to overcome these limitations by using sensors, a microcontroller, and an automatic water pump control system.

The system continuously monitors the soil moisture level using a soil moisture sensor. Additional sensors such as temperature, humidity, and water-level sensors can be used to monitor environmental conditions and the availability of water. The sensor readings are processed by a microcontroller such as an Arduino or ESP8266. Based on predefined moisture limits, the controller decides whether irrigation is required.

When the soil becomes dry and the moisture level falls below the required threshold, the microcontroller activates a relay module, which switches ON the water pump. Water is then supplied to the crops through a drip or sprinkler irrigation system. Once the soil reaches the desired moisture level, the controller automatically switches OFF the pump. This ensures that crops receive an appropriate amount of water without unnecessary consumption.

The system reduces manual intervention, water wastage, and irrigation time while improving the efficiency of agricultural water management. It can also be expanded with IoT technology to enable remote monitoring and control through a mobile application or web platform. Overall, the project provides a low-cost, reliable, and sustainable solution for smart agriculture.

Main Objectives

1. To develop an automatic irrigation system that controls water supply based on soil conditions.
2. To monitor soil moisture continuously using suitable sensors.
3. To reduce water wastage by supplying water only when required.
4. To automate water pump operation using a microcontroller and relay module.
5. To minimize manual effort involved in conventional irrigation.
6. To prevent over-irrigation and under-irrigation and maintain suitable soil moisture.
7. To improve water utilization efficiency and support sustainable agricultural practices.

2. Problem Statement

Problem Statement

Traditional agricultural irrigation systems often rely on manual operation and fixed watering schedules, which do not consider the actual moisture requirements of the soil. This can lead to

over-irrigation, under-irrigation, water wastage, and increased labor requirements. Farmers may also find it difficult to continuously monitor soil moisture and environmental conditions, especially in large agricultural fields.

Excessive irrigation can cause waterlogging, nutrient loss, and poor crop growth, while insufficient irrigation can result in soil dryness, plant stress, and reduced crop yield. In addition, manually switching pumps ON and OFF can consume time and may result in unnecessary water and energy consumption.

Therefore, there is a need for an automated and intelligent irrigation system that continuously monitors soil moisture and environmental conditions and supplies water only when required. The proposed Smart Irrigation and Water Management System addresses this problem by using sensors, a microcontroller, and an automatic water pump control mechanism to achieve efficient water utilization and reliable irrigation.

3. Proposed Solution

The proposed Smart Irrigation and Water Management System is an automated irrigation system designed to supply water to crops according to the actual moisture requirement of the soil. The system continuously monitors the soil moisture and environmental conditions using appropriate sensors and uses the collected information to control the irrigation process automatically.

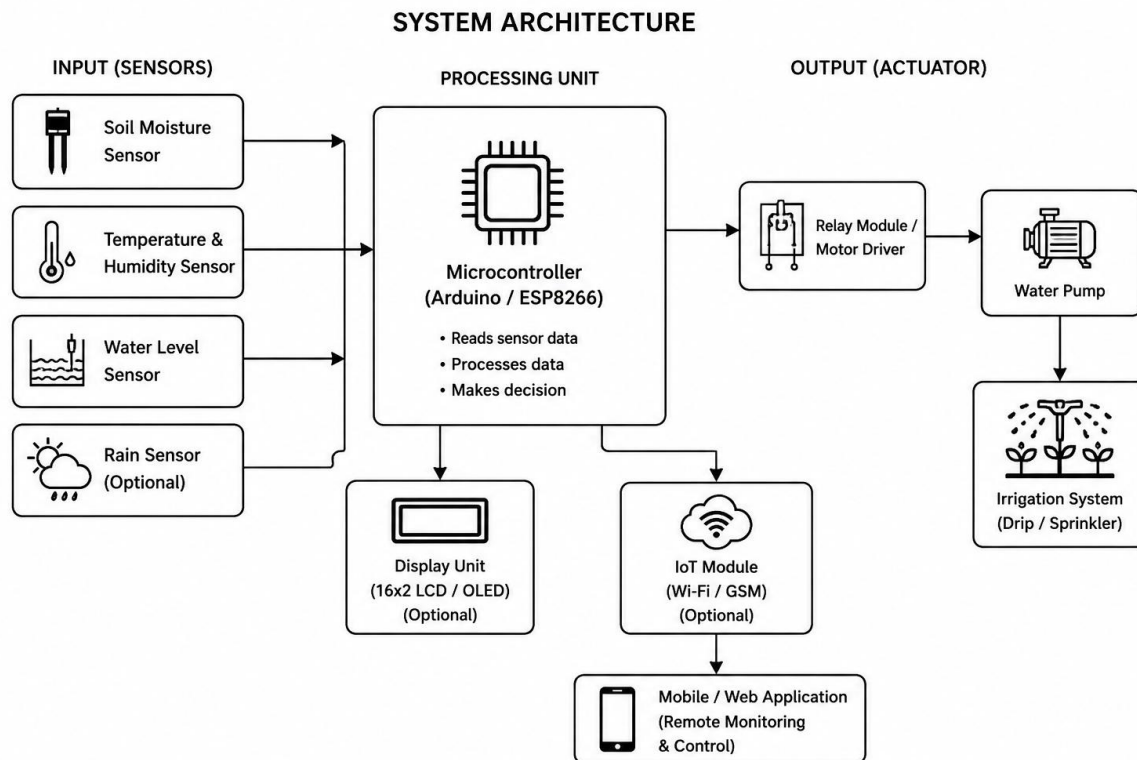
A soil moisture sensor is installed in the agricultural field to measure the moisture content of the soil. Additional sensors such as temperature and humidity sensors can be used to monitor environmental conditions, while a water-level sensor can be used to check the availability of water in the storage tank. All sensor readings are sent to a microcontroller such as Arduino or ESP8266 for processing.

The microcontroller compares the measured soil moisture value with a predefined threshold value. When the soil becomes dry and the moisture level falls below the required limit, the controller activates a relay module, which switches ON the water pump. The pump supplies water to the crops through a drip irrigation or sprinkler system. When the soil reaches the desired moisture level, the controller automatically switches OFF the pump.

The system can also be designed to prevent the pump from operating when the water level in the tank is too low. This protects the pump and avoids unnecessary energy consumption. With the addition of IoT connectivity, farmers can remotely monitor soil moisture, water level, temperature, humidity, and pump status using a mobile application or web interface.

The proposed system provides an efficient and reliable method of irrigation by supplying water only when it is required. It reduces water wastage, minimizes manual intervention, saves energy, and helps maintain suitable soil moisture for healthy crop growth. Therefore, the system offers a low-cost, intelligent, and sustainable solution for modern agricultural water management.

4. System Architecture



5. Circuit Table

Component	Pin	ESP8266 NodeMCU
Soil Moisture Sensor	VCC	3V3
Soil Moisture Sensor	GND	GND
Soil Moisture Sensor	AO	A0
DHT11	VCC	3V3
DHT11	GND	GND
DHT11	DATA	D2
Relay Module	VCC	VIN
Relay Module	GND	GND
Relay Module	IN	D3
Relay Module	COM	Pump power supply +
Relay Module	NO	Pump +
Water Pump	—	Pump power supply —

Power Supply

The ESP8266 can be powered through USB. The relay is powered from the appropriate 5V supply through VIN when the board is USB powered.

The water pump should use a suitable power supply according to its rated voltage and current. For a 5V pump, a regulated 5V supply or suitable boost converter should be used.

6. Circuit Design

The circuit consists of five major sections:

6.1 ESP8266 NodeMCU

The ESP8266 is the main controller. It receives sensor data and controls the relay according to the soil moisture condition.

6.2 Soil Moisture Sensor

The soil moisture sensor is connected to the analog input A0.

It provides an analog value representing the moisture condition of the soil.

Soil Sensor AO → ESP8266 A0

6.3 DHT11 Sensor

The DHT11 measures:

- Temperature.
- Relative humidity.

DHT11 DATA → ESP8266 D2

6.4 Relay Module

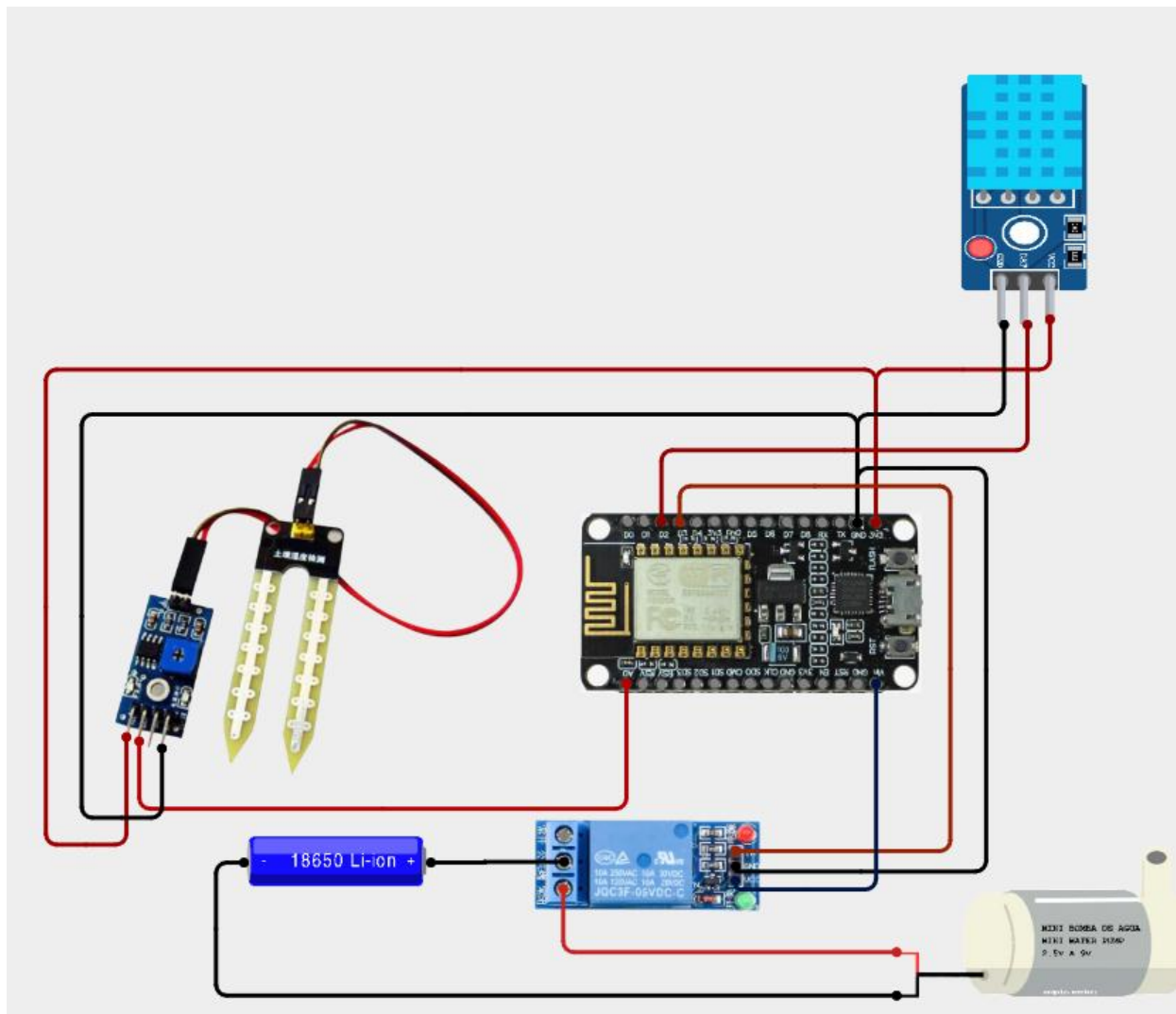
The relay provides electrical switching between the low-power controller and the water pump.

Relay IN → ESP8266 D3

6.5 Water Pump

The water pump supplies water to the plants. It is controlled through the relay and should not be connected directly to an ESP8266 GPIO pin.

CIRCUIT DESIGN:



7. Algorithm

Algorithm

1. Start the system.
2. Initialize the ESP8266.
3. Initialize the soil moisture sensor.
4. Initialize the DHT11 sensor.
5. Configure the relay as an output.
6. Read the soil moisture value.
7. Read temperature and humidity.
8. Compare the soil moisture value with the predefined threshold.
9. If the soil is dry, turn ON the relay.
10. The relay turns ON the water pump.

11. Continue monitoring the soil.
12. When sufficient moisture is reached, turn OFF the relay.
13. The water pump stops.
14. Display or transmit the sensor readings.
15. Repeat the process continuously.

8. Coding

The following Arduino program is designed for the ESP8266 NodeMCU used in the project.

```
#include <DHT.h>

#define SOIL_PIN A0
#define DHT_PIN D2
#define RELAY_PIN D3

#define DHT_TYPE DHT11

DHT dht(DHT_PIN, DHT_TYPE);

int dryThreshold = 700;
int wetThreshold = 500;

void setup() {

  Serial.begin(115200);

  pinMode(SOIL_PIN, INPUT);
  pinMode(RELAY_PIN, OUTPUT);

  // Pump OFF initially
```

```
digitalWrite(RELAY_PIN, HIGH);

dht.begin();

Serial.println("Smart Irrigation System Started");
}

void loop() {

    int soilValue = analogRead(SOIL_PIN);

    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    Serial.println("-----");

    Serial.print("Soil Moisture: ");
    Serial.println(soilValue);

    if (isnan(temperature) || isnan(humidity)) {

        Serial.println("DHT11 reading failed");

    } else {

        Serial.print("Temperature: ");
        Serial.print(temperature);
        Serial.println(" C");
    }
}
```



```
Serial.print("Humidity: ");  
Serial.print(humidity);  
Serial.println(" %");  
}  
  
if (soilValue >= dryThreshold) {  
  
    digitalWrite(RELAY_PIN, LOW);  
  
    Serial.println("Soil Status: DRY");  
    Serial.println("Pump Status: ON");  
  
}  
  
else if (soilValue <= wetThreshold) {  
  
    digitalWrite(RELAY_PIN, HIGH);  
  
    Serial.println("Soil Status: WET");  
    Serial.println("Pump Status: OFF");  
  
}  
  
else {  
  
    Serial.println("Soil Status: MODERATE");  
    Serial.println("Pump Status: Previous State");  
  
}
```

```
delay(2000);  
}
```

Code Explanation

- A0 reads the soil moisture sensor.
- D2 receives data from the DHT11.
- D3 controls the relay.
- dryThreshold determines when irrigation should start.
- wetThreshold determines when irrigation should stop.
- The DHT11 provides temperature and humidity information.
- The relay controls the water pump.

The threshold values should be calibrated according to the actual soil and sensor used.

9. System Working

When the system is powered ON, the ESP8266 initializes all connected sensors and the relay.

The soil moisture sensor continuously checks the moisture condition of the soil.

Dry Soil Condition

When the soil moisture value indicates that the soil is dry:

Soil Dry



ESP8266 detects dry condition



Relay ON



Water Pump ON



Water supplied to plants

Wet Soil Condition

When the soil reaches the required moisture level:

Soil Wet



ESP8266 detects sufficient moisture



Relay OFF



Water Pump OFF

At the same time, the DHT11 continuously measures temperature and humidity.

This process repeats automatically, allowing the system to maintain suitable soil moisture.

10. Bill of Materials

S.No.	Component	Quantity
1	ESP8266 NodeMCU	1
2	Soil Moisture Sensor	1
3	1-Channel Relay Module	1
4	Mini Water Pump	1
5	DHT11 Sensor	1
6	Breadboard	1
7	Jumper Wires	1 set
8	Suitable Pump Power Supply	1

Software Requirements

- Arduino IDE
- ESP8266 Board Package
- DHT Sensor Library
- Circuit Designer for circuit design/simulation

11. Prototype Implementation

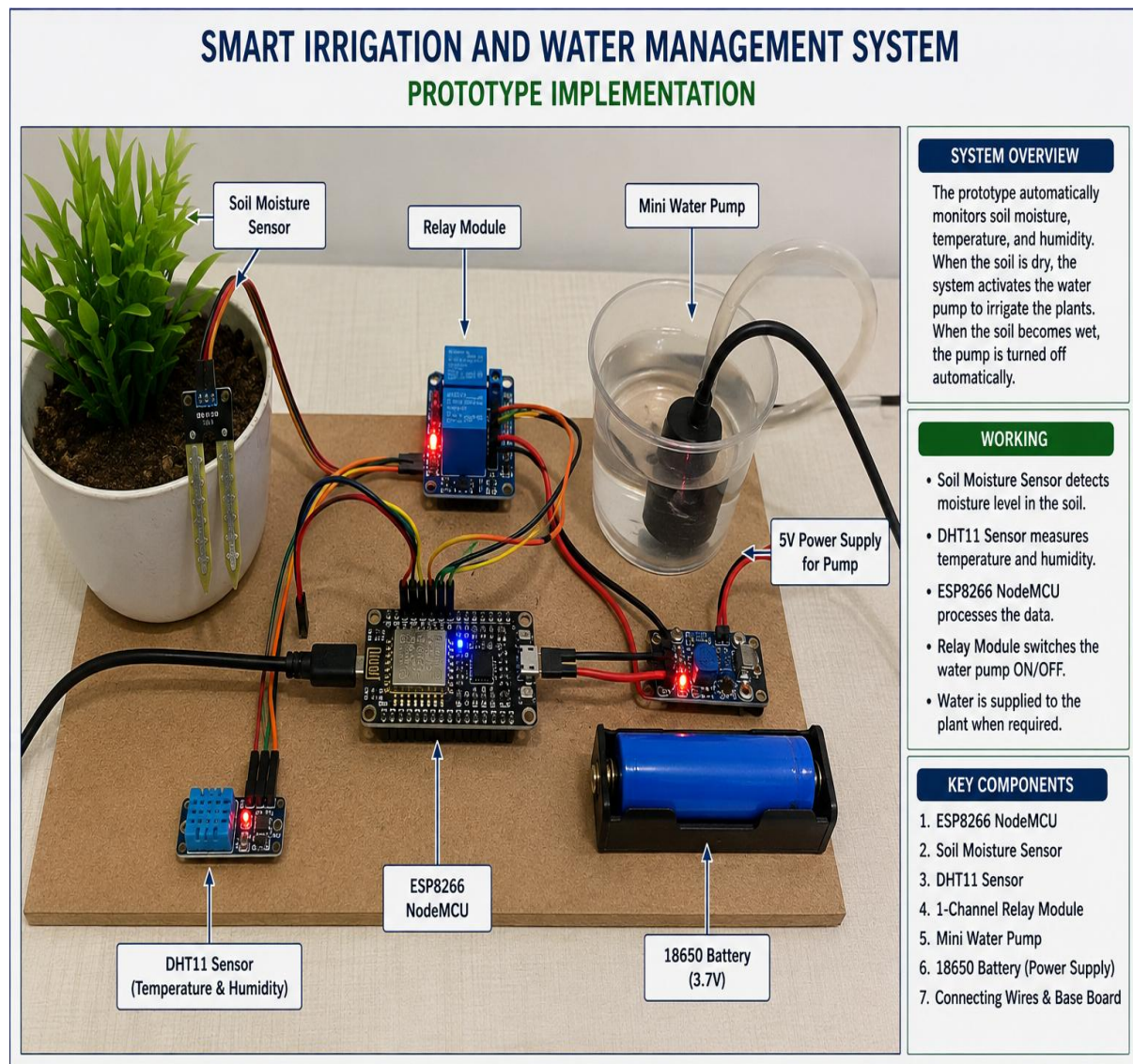
The prototype was designed using an ESP8266 NodeMCU, soil moisture sensor, DHT11 sensor, relay module and mini water pump.

The components were first arranged and connected in Circuit Designer to develop the circuit design.

The ESP8266 was selected as the main controller because it provides sufficient GPIO and analog input capability and can also support Wi-Fi-based IoT extensions.

The prototype operates automatically according to the soil moisture condition.

Prototype Process



Simulation

The circuit can be represented in Circuit Designer. For simulation, a supported simulation microcontroller such as Arduino Uno can be used if the ESP8266 NodeMCU configuration does not provide an active simulation/verification environment.

12. Results & Performance Analysis

Expected Results

The system successfully demonstrates automatic irrigation based on soil moisture.

Condition	Sensor Condition	Relay	Pump
Dry soil	Moisture below required level	ON	ON
Moderate soil	Intermediate value	Maintains previous state	Maintains previous state
Wet soil	Moisture sufficient	OFF	OFF

Performance

The system provides the following benefits:

- Automatic irrigation.
- Reduced unnecessary water usage.
- Reduced manual intervention.
- Prevention of over-irrigation.
- Continuous environmental monitoring.
- Automatic pump control.
- Potential for remote IoT monitoring

Conclusion

The Smart Irrigation and Water Management System provides an efficient approach to automated agricultural irrigation. By continuously monitoring soil moisture and environmental conditions, the system can automatically operate the water pump only when irrigation is required.

The ESP8266 acts as the central controller, while the soil moisture sensor provides information about the condition of the soil. The DHT11 provides temperature and humidity measurements, and the relay controls the water pump.

The proposed system can help reduce water wastage and improve irrigation efficiency. In future development, additional features such as water-level monitoring, flow measurement, mobile applications, cloud data logging, weather-based irrigation and solar-powered operation can be incorporated.